Mathematical background

Christos Dimitrakakis

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Outline

Elementary background

- Set theory and logic
- Functions, sequences and sums

Probability background

- Probability facts
- Conditional probability and independence
- Random variables, expectation and variance

Linear algebra

- Vectors
- Norms and distances
- Linear operators and matrices

Calculus

- Univariate caclulus
- Multivariate calculus

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Set theory

- First, consider some universal set Ω and the empty set $\emptyset$
- A set A is a collection of points x in Ω.
- $\{x \in \Omega : f(x)\}$: the set of points in Ω for which $f(x)$ is true.

Unary operators

- Negation: $\neg A = \{x \in \Omega : x \notin A\} = \Omega \setminus A$.

Binary operators

- Union: $A \cup B = \{x \in \Omega : x \in A \vee x \in B\}$ - (c.f. $A \vee B$)
- Intersection: $A \cap B = \{x \in \Omega : x \in A \wedge x \in B\}$ - (c.f. $A \wedge B$)
- Difference: $A \setminus B = \{x \in \Omega : x \in A \wedge x \notin B\}$ (c.f. $A \wedge \not B$)

Binary relations

- $A \subset B$ if $x \in A \Rightarrow x \in B$ - (c.f. $A \implies B$)
- $A = B$ if $x \in A \Leftrightarrow x \in B$ - (c.f. $A \Leftrightarrow B$)

Logic

Statements

- A statement A may be true or false

Unary operators

- negation: $\neg A$ is true if A is false (and vice-versa).

Binary operators

- or: $A \vee B$ (A or B) is true if either A or B are true.
- and: $A \wedge B$ is true if both A and B are true.
- implies: $A \Rightarrow B$: is false if A is true and B is false.
- iff: $A \Leftrightarrow B$: is true if A, B have equal truth values.

Operator precedence

$\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$

Functions

Definition

A function $f : X \to Y$ is a relation from a set X to a set Y so that, for any $x \in X$, there exists a unique $y \in Y$ so that $y = f(x)$.

Inverse of a function

A function $f : X \to Y$ has an inverse $f^{-1} : Y \to X$ if, for any $y \in Y$, there exists a unique $x \in X$ such that $f(x) = y$, and so that $f^{-1}(y) = x$.

Sequences

Sequences

A sequence $x_1, \ldots, x_t$ is written $\{x_i | i = 1, \ldots, t\}$ or $\{x_i\}$ for brevity

Sums

A sum of a sequence of variables $\{x_i\}$ is defined as:

$$\sum_{i=1}^{t} x_i = x_1 + x_2 + \cdots + x_t.$$

The python notation `sum([f(x) for x in A])` corresponds to:

$$\sum_{x \in A} f(x)$$

Products

A product is similarly defined at

$$\prod_{i=1}^{t} x_i = x_1 \times x_2 \times \cdots \times x_t.$$

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Events as sets

The universe and random outcomes

- The Ω contains all events that can happen.
- When something happens, we observe an element $\omega \in \Omega$.

Events in the universe

- An event is true if $\omega \in A$, and false if $\omega \notin A$.
- The negative event $\neg A$ is the set $\Omega \setminus A$
- The possible events are a collection of subsets Σ of Ω so that

(i) $\Omega \in \Sigma$, (ii) $A, B \in \Sigma \Rightarrow A \cup B \in \Sigma$ (iii) $A \in \Sigma \Rightarrow \neg A \in \Sigma$

Example: Traffic violation

- A car is moving with speed $\omega \in [0, \infty)$ in front of the speed camera.
- $A_0 = [0, 50]$: below the speed limit
- $A_1 = (50, 60]$: low fine
- $A_2 = (60, \infty]$: high fine
- $A_3 = (100, \infty)$: Suspension of license
- All combinations of the above events are interesting.

Probability fundamentals

Probability measure P

Probability can be seen as an area-like function assigning a likelihood to sets.

- Defined on a universe Ω
- $P : \Sigma \to [0, 1]$ is a function of subsets (Σ is the collection of subsets) of Ω.
- A subset $A \subset \Omega$ is an event and P measures its likelihood.

Axioms of probability

- $P(\Omega) = 1$
- For all $A \subseteq \Omega$, $P(A) \geq 0$.
- For $A, B \subseteq \Omega$, if $A \cap B = \emptyset$ then $P(A \cup B) = P(A) + P(B)$.

Probability tricks

Implication

If $A \subseteq B$ then $P(A) \leq P(B)$.

Partition

$\{A_i\}$ is a partition of Ω if $A_i \cap A_j = \emptyset \ \forall i \neq j$ and $\bigcup_{i=1}^{n} A_i = \Omega$. A partition of Ω defines a complete set of mutually exclusive alternatives.

Marginalisation

Let $A_1, \ldots, A_n \subset \Omega$ be a partition of Ω. Then

$$P(B) = \sum_{i=1}^{n} P(B \cap A_i).$$

Conditional probability

Definition (Conditional probability)

The conditional probability of an event A given an event B is defined as

$$P(A|B) \triangleq \frac{P(A \cap B)}{P(B)}$$

The above definition requires $P(B)$ to exist and be positive.

Conditional probabilities as a collection of probabilities

More generally, we can define conditional probabilities as simply a collection of probability distributions:

$$\{P_\theta(A) \mid \theta \in \Theta\},$$

where Θ is an arbitrary set.

The theorem of Bayes

Theorem (Bayes's theorem)

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

The theorem of Bayes

Theorem (Bayes's theorem)

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The general case

If $A_1, \ldots, A_n$ are a partition of Ω, meaning that they are mutually exclusive events (i.e. $A_i \cap A_j = \emptyset$ for $i \neq j$) such that one of them must be true (i.e. $\bigcup_{i=1}^{n} A_i = \Omega$), then

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$$

and

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)}$$

Independence

Independent events

A, B are independent iff $P(A \cap B) = P(A)P(B)$.

Conditional independence

A, B are conditionally independent given C iff $P(A \cap B|C) = P(A|C)P(B|C)$.

Random variables

A random variable $f : \Omega \to \mathbb{R}$ is a real-value function measurable with respect to the underlying probability measure P, and we write $f \sim P$.

The distribution of f

The probability that f lies in some subset $A \subset \mathbb{R}$ is

$$P_f(A) \triangleq P(\{\omega \in \Omega : f(\omega) \in A\}).$$

Independence

Two RVs f, g are independent in the same way that events are independent:

$$P(f \in A \wedge g \in B) = P(f \in A)P(g \in B) = P_f(A)P_g(B).$$

In that sense, $f \sim P_f$ and $g \sim P_g$.

IID (Independent and Identically Distributed) random variables

A sequence x_t of r.v.s is IID if $x_t \sim P$ $(x_1, \ldots, x_t, \ldots, x_T) \sim P^T$.

Expectation

For any real-valued random variable $f : \Omega \to \mathbb{R}$, the expectation with respect to a discrete probability measure P is

$$\mathbb{E}_P(f) \triangleq \sum_{\omega \in \Omega} f(\omega) P(\omega).$$

Linearity of expectations

$$\mathbb{E}_P(x + y) = \mathbb{E}_P(x) + \mathbb{E}_P(y) \qquad \text{for any RVs } x, y.$$

Independence

If x, y are independent RVs then $\mathbb{E}_P(xy) = \mathbb{E}(x)\,\mathbb{E}(y)$.

Correlation

If x, y are not correlated then $\mathbb{E}_P(xy) = \mathbb{E}(x)\,\mathbb{E}(y)$.

Expectation for general Ω

Ω is not always finite (or even discrete). When Ω is continuous, we can instead define the density p so that

$$\mathbb{E}_P(f) \triangleq \int_\Omega f(\omega)p(\omega)d\omega.$$

The density p is related to the probability P as follows:

- for any subset $A \subset \Omega$, $P(A) = \int_A p(\omega)d\omega$.

In mathematical statistics and probability, people prefer using a so-called probability measure P, which allows us to use the following notation no matter whether Ω is discrete, continuous, or something else.

$$\mathbb{E}_P(f) \triangleq \int_\Omega f(\omega)dP(\omega).$$

Measure theory is an advanced topic, but if you see this notation, it means something similar to the standard integral and sum (see the Calculus section).

Conditional expectation

Independence

If x, y are independent RVs then they are also uncorrelated (but not vice-versa)

Conditional expectation of f given some event B

The conditional expectation of a random variable $f : \Omega \to \mathbb{R}$, with respect to a probability measure P conditioned on some event B is simply

$$\mathbb{E}_P(f|B) = \sum_{\omega \in \Omega} f(\omega) P(\omega|B).$$

Conditional expectatio of f wrt RV g.

This uses a similar notation, but the meaning is different. It is not a single number, but a function of what value g takes:

$$\mathbb{E}_P(f|g = y) = \sum_{\omega \in \Omega} f(\omega) P(\omega|B), \qquad B = \{\omega \ | \omega = y\}$$

Variance

For any real-valued random variable $f : \Omega \to \mathbb{R}$, the variance with respect to a probability measure P is

$$\mathbb{V}_P(f) = \sum_{\omega \in \Omega} [f(\omega) - \mathbb{E}_P(f(\omega)]^2 P(\omega).$$

Linearity of variance

If f, g are uncorrelated RVs

$$\mathbb{V}_P(f + g) = \mathbb{V}_P(f) + \mathbb{V}_P(g).$$

Variance products

If f, g are independent RVs

$$\mathbb{V}_P(f + g) = \mathbb{E}_P(f)^2 \, \mathbb{V}_P(g) + \mathbb{E}_P(g)^2 \, \mathbb{V}_P(f) + \mathbb{V}_P(f) \, \mathbb{V}_P(g).$$

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Vector space F

- The Euclidean space $\mathbb{R}^d$ is a vector space.
- Vector spaces have the following properties.

Axioms of F

Vector space F axioms

Here we consider a vector space F. Typically, it is a subset of the Euclidean d-dimensional space, ie. $F \subset \mathbb{R}^d$.

- $(x + y) + z = x + (y + z)$, for all $x, y, z \in F$.
- $x + y = y + x$, for all $x, y \in F$.
- There is a zero element $0 \in F$ such that $x + 0 = x$ for all $x \in F$.
- For all $x \in F$, there is an element $-x \in F$ so that $x + (-x) = 0$.
- $a(x + y) = ax + ay$, For any $a \in \mathbb{R}$, $x, y \in F$.
- $(a + b)x = ax + bx$, For any $a, b \in \mathbb{R}$, $x \in F$.

The real vector space $F = \mathbb{R}^d$

For $a \in \mathbb{R}$ and $x, y \in F$,

- Elements: $x = (x_1, \ldots, x_d)$, $y = (y_1, \ldots, y_d)$
- Addition: $x + y = (x_1 + y_1, \ldots, x_d + y_d)$.
- Scalar multiplication: $ax = (ax_1, \ldots, ax_d)$.
- Negation: $-x = (-1)x$.
- Zero element: $0 = (0, \ldots, 0)$

Norms

A norm $\|\cdot\|$ is a scalar operator on a vector with the properties

- $\|x\| \geq 0$ with equality iff $x = 0$.
- $\|x + y\| \leq \|x\| + \|y\|$
- $\|cx\| = c\|x\|$ for every scalar c.

Example: Manhattan norm $\mathbb{R}^n$

$\|x\|_1 = \sum_{i=1}^{n} |x_i|$

Example: Euclidean norm in $\mathbb{R}^n$

$\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i}$

Example: p-norm

$\|x\|_p = \left(\sum_{i=1}^{n} x_i\right)^{1/p}$

Metrics

A metric (or distance) on $\mathcal{X}$ is a function $d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ with the properties:

- $d(x, y) \geq 0$ with equality iff $x = y$.
- $d(x, y) = d(y, x)$.
- $d(x, y) \leq d(x, z) + d(z, y)$.

Example: Euclidean distance

$$d_2(x, y) = \|x - y\|_2$$

Linear operators

Linear operator $A : F \to G$.

A linear operator A from one vector space F to another, G has the following properties:

- $A(x + y) = Ax + Ay$ for all $x, y \in F$.
- $A(ax) = a(Ax)$, for all $a \in \mathbb{R}$, $x \in F$.

Matrices in $\mathbb{R}^{n \times m}$.

A matrix $\boldsymbol{A} \in \mathbb{R}^{n \times m}$ is a tabular array

$$\boldsymbol{A} = \begin{bmatrix} A_{1,1} & \cdots & A_{1,m} \\ \vdots & \ddots & \vdots \\ A_{n,1} & \cdots & A_{n,m} \end{bmatrix}$$

Matrices can be seen as linear operators $\boldsymbol{A} : \mathbb{R}^n \to \mathbb{R}^m$, mapping vectors in $\mathbb{R}^n$ to vectors in $\mathbb{R}^m$.

Multiplication operators

Matrix multiplication

For $\boldsymbol{A} \in \mathbb{R}^{n \times d}$, $\boldsymbol{B} \in \mathbb{R}^{d \times m}$, the ij-th element of the result of the multiplication $\boldsymbol{AB}$ is

$$(\boldsymbol{AB})_{i,j} = \sum_{k=1}^{d} A_{i,k} B_{k,j}.$$

so that $\boldsymbol{AB} \in \mathbb{R}^{n \times m}$.

Matrix-vector multiplication

A matrix $\boldsymbol{A} \in \mathbb{R}^{n \times m}$ defines the following linear operator $\boldsymbol{A} : \mathbb{R}^m \to \mathbb{R}^n$.

$$\boldsymbol{Ax} = \left(\sum_{j=1}^{m} A_{i,j} x_j : i = 1, \ldots, n \right)$$

All vectors $\boldsymbol{x} \in \mathbb{R}^m$ are equivalent to matrices in $\mathbb{R}^{m \times 1}$.

Matrix inverses

The identity matrix $\boldsymbol{I} \in \mathbb{R}^{n \times n}$

- For this matrix, $I_{i,i} = 1$ and $I_{i,j} = 0$ when $j \neq i$.
- $\boldsymbol{I}\boldsymbol{x} = \boldsymbol{x}$ and $\boldsymbol{I}\boldsymbol{A} = \boldsymbol{A}$.

The inverse of a matrix $\boldsymbol{A} \in \mathbb{R}^{n \times n}$

$\boldsymbol{A}^{-1}$ is called the inverse of A if

- $\boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}$.
- or equivalently $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$.

The pseudo-inverse of a matrix $\boldsymbol{A} \in \mathbb{R}^{n \times m}$

- $\tilde{\boldsymbol{A}}^{-1}$ is called the left pseudoinverse of $\boldsymbol{A}$ if $\tilde{\boldsymbol{A}}^{-1}\boldsymbol{A} = \boldsymbol{I}$.
- $\tilde{\boldsymbol{A}}^{-1}$ is called the right pseudoinverse of $\boldsymbol{A}$ if $\boldsymbol{A}\tilde{\boldsymbol{A}}^{-1} = I$.

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Derivatives

Derivative

The derivative of a single-argument function is defined as:

$$\frac{d}{dx} f(x) = \lim_{\epsilon \to 0} \frac{f(x+\epsilon) - f(x)}{\epsilon}.$$

f must be absolutely continuous at x for the derivative to exist.

Subdifferential

For non-differential functions, we can sometimes define the set of all subderivatives:

$$\partial f(x) = [\lim_{\epsilon \to 0} \frac{f(x) - f(x-\epsilon)}{\epsilon}, \lim_{\epsilon \to 0} \frac{f(x+\epsilon) - f(x)}{\epsilon}]$$

Elementary differentiation

The chain rule

The composition fg of two real-valued functions: $f(y), g(x)$, is:

$$fg(x) = f[g(x)], \qquad y \triangleq g(x), z \triangleq f(y).$$

The chain rule of differentiation states that:

$$\frac{df}{dx} = \frac{df}{dg}\frac{dg}{dx}, \qquad \frac{df[g(x)]}{dx} = \frac{df(y)}{dy}\frac{dg(x)}{dx} \qquad \frac{dz}{dx} = \frac{dz}{dy}\frac{dy}{dx},$$

given in three variant notations.

Common derivatives

- $\frac{d}{dx}[cf] = c\frac{df}{dx}$ for any constant $c \in \mathbb{R}$.
- $\frac{d}{dx}[f+g] = \frac{df}{dx} + \frac{dg}{dx}$
- $\frac{d}{dx}[fg] = \frac{df}{dx}g + f\frac{dg}{dx}$.
- $\frac{d}{dx}x^n = nx^{n-1}$.
- $\frac{d}{dx}(1/x) = \ln(x)$.
- $\frac{d}{dx}(e^x) = e^x$.

Exercises

Simple Derivatives

Show that $\frac{d}{dx}2^x = 2_2x$, using the fact that $c^x = e^{x \ln c}$.

Squared error

For some sequence $x_1, \ldots, x_T$ of bounded real numbers, let

$$f(\theta) \triangleq -\left[\frac{1}{T}\sum_{t=1}^{T}(x_t - \theta)^2\right]$$

be the mean-squared error. To minimise the squared error, we need to find θ such that $\frac{d}{d\theta}f(\theta) = 0$, if the second derivative is positive. Show that this equals $\theta^* = \frac{1}{T}\sum_{t=1}^{T} x_t$. Bonus: show the second derivative $\frac{d^2}{d\theta^2}f(\theta)$ is positive.

Integrals

Riemann integral

The Reimann integral is obtained by taking a horizontal discretisation of a function to the limit:

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{t=1}^{n} f(x_t)\frac{b-a}{n}, \qquad x_t = a + (t-1)\cdot\frac{b-a}{n}$$

Because the Reimann integral does not always exist, it is more convenient to use this more advanced concept:

Lebesgue integral

This integral is obtained by taking a vertical discretisation of a function to the limit. Let λ be the Lebesgue measure (i.e. area) of a set. Then:

$$\int_X f(x)d\lambda(x) = \lim_{n\to\infty} \sum_{t=1}^{n} y_t \lambda(S_t),$$

where $S_t = \{x : f(x) \in (y_{t-1}, y_t]\}$, are the "slices" of y values, with $y_0 = -\infty$, $y_n = \sup_x f(x)$.

Fundamental theorem of calculus

$$f(x) = \frac{d}{dx} \int_a^x f(t)dt$$

If $\frac{d}{dx}F = f$ then its integral from a to b is:

$$\int_a^b f(x)dx = F(b) - F(a),$$

Multivariate Functions

We consider functions operating in multi-dimensional Euclidean spaces, i.e. with multiple inputs and/or multiple outputs.

$f : \mathbb{R}^n \to \mathbb{R}$.

These are functions with multiple arguments, and a single output.

- Any $x \in \mathbb{R}^n$ is $x = (x_1, \ldots, x_n)$, with $x_i \in \mathbb{R}$.
- We write $f(x)$ instead of $f(x_1, \ldots, x_n)$.

$f : \mathbb{R}^n \to \mathbb{R}^m$.

These functions have multiple inputs and outputs.

- If $y = f(x)$ then y_i is the i-th component of $y \in \mathbb{R}^m$.
- Can be seen as m functions $f_i : \mathbb{R}^n \to \mathbb{R}$, with $y_i = f_i(x)$.

Derivatives in many dimensions

Partial derivative

The partial derivative of $f : \mathbb{R}^n \to \mathbb{R}$ with respect to its i-th argument is: $\frac{\partial}{\partial x_i} f(x)$, where we see all x_j with $j \neq i$ as fixed.

Gradient of f

This is the column vector of all its partial derivatives:

$$\nabla_x f(x) = \left(\frac{\partial}{\partial x_1} f(x), \cdots \frac{\partial}{\partial x_i} f(x), \cdots \frac{\partial}{\partial x_n} f(x) \right)^\top$$

When $f : \mathbb{R}^n \to \mathbb{R}^m$, the gradient is an $n \times m$ matrix called the Jacobian.
Some textbooks define the Jacobian as an $m \times n$ matrix instead. This changes how the chain rule applies

Directional derivative

We can also define the derivative with respect to a direction $\delta \in \mathbb{R}^n$:

$$D_\delta f(x) = \lim_{\epsilon \to 0} \frac{f(x + \epsilon\delta) - f(x)}{\epsilon}.$$

For simplicity, $\|\delta\| = 1$.

Matrix calculus

Here we review some basic facts for matrix calculus.

- Gradient: If $f : \mathbb{R}^n \to \mathbb{R}^m$, then $\nabla f \in \mathbb{R}^{n \times m}$.
- Chain rule: If $g : \mathbb{R}^k \to \mathbb{R}^n$, so that $f \circ g : \mathbb{R}^k \to \mathbb{R}^m$, i.e. $x = g(z)$, $y = f(x)$, $x \in \mathbb{R}^n$, $z \in \mathbb{R}^k$, $y \in \mathbb{R}^m$:

$$\nabla[f \circ g] = \nabla g \nabla f, \qquad \text{or} \ , \quad \nabla_z[f(g(z))] = \nabla_z g(z) \nabla_x f(x)$$

with $\nabla[f \circ g] \in \mathbb{R}^{k \times m}$.
From this, the following facts follow:

- If $g(z) = Az$, with $A \in \mathbb{R}^{n \times k}$, then $g : \mathbb{R}^k \to \mathbb{R}^n$. Applying the chain rule:

$$\nabla_z[f(Az)] = A^\top \nabla f(Az),$$

where $\nabla f(Az) = \nabla_x f(x)$ for $x = Az$.

- We also have $\nabla(v^\top v) = 2\nabla v v$.

If we use the definition of the Jacobian with $\nabla f \in \mathbb{R}^{m \times n}$ then the chain rule order changes to $\nabla[f \circ g] = \nabla f \nabla g$, so be careful!

Multi-dinemsional integrals

We only rarely use explicit such integrals:

$$\int_{X_1} \int_{X_2} f(x_1, x)\, dx_1\, dx_2, \qquad x_1 \in X_1, x_2 \in X_2$$

For a general integral over multiple dimensions, we will use Lebesgue integrals:

$$\int_X f(x) d\lambda(x), \qquad x \in X,$$

where e.g. $X = X_1 \times X_2$ so that $x = (x_1, x_2)$. This avoids having multiple $\int$ and dx_i symbols.
Lebesgue notation is particularly useful with distributions. Then, for example, the expectation of some rv $f : X \to \mathbb{R}$ with respect to a probability measure P is:

$$\mathbb{E}_P[f] = \int_X f(x) dP(x),$$

and it does not matter if X is a discrete or continuous set.